Standard Operating Procedure (SOP)

Title:

Active Directory User Account Lockout Resolution

Document Version / Revision:

Version 1.0

Date of Issue / Revision:

Author / Owner:

Nabin Shrestha

Department / Team:

IT Operations / Network & Systems Administration

2. Purpose

The purpose of this Standard Operating Procedure (SOP) is to provide IT staff with a **clear, consistent, and standardized process** to identify, investigate, and resolve **Active Directory user account lockouts**.

This SOP ensures that:

- Locked user accounts are **promptly identified** and resolved to minimize downtime.
- The **root cause of account lockouts** is accurately determined to prevent recurring issues.
- Actions are performed in a **secure, auditable, and compliant manner**, adhering to organizational policies and industry standards (e.g., ISO/IEC 27001).
- IT staff follow **consistent procedures** to maintain system integrity, avoid unauthorized access, and ensure business continuity.

By following this SOP, the organization can maintain **efficient user account management**, reduce the risk of security incidents, and improve overall **network and system reliability**.

3. Scope

This SOP applies to all IT staff and system administrators responsible for managing **Active Directory (AD) user accounts** within the organization.

Included in this SOP:

- Identification of **locked Active Directory user accounts**.
- Investigation of the **root cause of account lockouts**.
- Procedures to **unlock user accounts** safely and securely.
- Documentation of lockout events and actions taken.
- Escalation procedures if standard resolution steps fail.

Excluded from this SOP:

- Password resets or recovery procedures for **non-AD systems** (e.g., email-only accounts, third-party applications).
- Issues unrelated to **user account lockouts**.
- General network or system troubleshooting not connected to AD lockouts.

This SOP ensures that **only account lockout events within Active Directory** are addressed, providing a clear boundary for IT staff responsibilities.

4. Roles and Responsibilities

This section defines the roles and responsibilities of personnel involved in managing and resolving Active Directory user account lockouts:

IT Helpdesk

- Monitor user-reported issues and system alerts for account lockouts.
- Log incidents in the ticketing system with relevant details (username, time, symptoms).
- Perform initial verification to confirm the account is locked.

System Administrator

- Investigate the source of the lockout using tools such as **Event Viewer** and **LockoutStatus**.
- Unlock user accounts and reset passwords as required.
- Ensure actions comply with **security policies** and **audit requirements**.
- Document all steps taken in the incident ticket.

Network / Security Team

- Monitor for patterns of repeated or suspicious lockouts that may indicate **security threats**.
- Escalate incidents to senior administrators or security officers if unusual activity is detected.
- Provide guidance on **preventive measures** to reduce recurring account lockouts.

All Roles

- Ensure communication is clear, timely, and documented for accountability.
- Follow **organizational security standards** and SOP guidelines at all times.

5. Prerequisites / Requirements

Before performing the Active Directory user account lockout resolution procedure, the following **access, tools, and knowledge** are required:

Access / Permissions:

- Domain Administrator or delegated rights to unlock user accounts.
- Read access to Event Logs on all relevant domain controllers.
- PowerShell administrative access to run Active Directory cmdlets.

Tools / Software Required:

- **Event Viewer** (Windows built-in).
- **LockoutStatus Tool** (Microsoft or third-party AD lockout tool).
- **Active Directory Module for Windows PowerShell**.
- Ticketing system for incident logging and tracking.
- Optional: Remote Desktop access to domain controllers if needed.

Knowledge / Skills:

- Understanding of **Windows Active Directory structure** (users, OUs, domain controllers).
- Familiarity with **Event Viewer** and **Event IDs**, specifically **4740** for account lockouts.
- Basic **PowerShell scripting** for AD operations.
- Understanding of **network authentication processes** (Kerberos/NTLM).
- Awareness of **security and audit policies** related to user account management.

Optional / Recommended:

- Knowledge of **common causes of repeated account lockouts** (bad passwords on other devices, service accounts).
- Familiarity with organizational **escalation policies** for recurring or suspicious lockouts.

6. Definitions / Acronyms

- **AD (Active Directory):** Microsoft directory service used to manage users, computers, and network resources.
- **OU (Organizational Unit):** A container within AD used to organize users, groups, and computers.
- **Event ID 4740:** Windows security event indicating a user account has been locked out.
- **Caller Computer:** The machine or device that triggered the account lockout.
- **DC (Domain Controller):** Server that authenticates users and enforces AD policies.
- **Kerberos / NTLM:** Authentication protocols used by AD for network login and security.
- **LockoutStatus Tool:** Utility to track locked accounts across domain controllers.
- **AD Module for PowerShell:** Windows module for managing Active Directory via PowerShell commands.
- **Ticketing System:** Software used to log, track, and resolve IT incidents.

7. Procedure / Step-by-Step Instructions

The following steps outline the standardized procedure for investigating and resolving Active Directory user account lockouts:

Step 1: Detect Locked Accounts

- Monitor **user-reported issues** and system alerts for account lockouts.
- Review security event logs on domain controllers using **Event Viewer**.
- Look for **Event ID 4740**, which indicates a locked-out user account.

Step 2: Identify Account and Source Computer (Caller Computer)

- Open Event Viewer → Windows Logs → Security → Filter by Event ID 4740.
- Check the **Target Account Name** to identify the locked user account.
- Check the **Caller Computer Name** field to determine which device triggered the lockout.

Step 3: Locate the Responsible Domain Controller

- Use the **LockoutStatus tool** to identify the domain controller with the latest lockout event.
- Confirm which DC is currently authenticating the user and generating the lockout.

Step 4: Verify Cause of Lockout

- Check for repeated failed login attempts from multiple devices (workstations, mobile devices, service accounts).
- Investigate scheduled tasks, mapped drives, or cached credentials that might trigger repeated lockouts.
- Determine whether the lockout is **legitimate** (user error) or **suspicious** (potential security issue).

Step 5: Unlock Account and Reset Password (if needed)

- Using **Active Directory Users and Computers** or **PowerShell**, unlock the user account.
- Reset the password if required, following security policies.
- Confirm the account is functional by having the user attempt to log in.

Step 6: Document the Incident

- Record all actions in the **ticketing system**, including:
- ✓ Locked user account name
- ✓ Caller computer
- ✓ Root cause of lockout
- ✓ Steps taken to resolve
- ✓ Any escalation performed
- Maintain records for **audit and compliance purposes**.

Step 7: Follow Escalation Procedure

- If the lockout persists or is suspected to be a security threat:
- ✓ Escalate to **senior system administrator or security team**.
- ✓ Provide all gathered logs and analysis.
- ✓ Follow **organization's escalation and incident response policies**.

Step 8 (Optional / Preventive Measures)

- Educate users about **common causes of repeated lockouts** (e.g., old credentials on mobile devices).
- Monitor for patterns of repeated lockouts to identify misconfigured systems or accounts.
- Update firewall, AD policies, or authentication settings if necessary.

8. Troubleshooting / Common Issues

This section provides guidance on resolving recurring or unexpected problems during Active Directory account lockout investigations.

1. Account Keeps Locking Repeatedly

- Check for **saved credentials** on user devices (laptops, mobile phones, tablets).
- Verify **service accounts or scheduled tasks** that may be using old passwords.
- Use **Event Viewer** to track repeated failed logins and correlate with devices.
- Educate the user on proper login procedures to prevent repeated lockouts.

2. Lockout Source is Unknown

- Use the **LockoutStatus tool** across all domain controllers to identify the responsible DC.
- Check for **replicated accounts** in multiple OUs that may trigger lockouts.
- Investigate possible **network authentication issues** (VPN, remote access, cached credentials).

3. Tools Fail to Connect to Domain Controller

- Verify network connectivity to the DC and proper **DNS resolution**.
- Ensure **Firewalls or security policies** are not blocking administrative access or PowerShell remoting.
- Confirm you have **appropriate administrative privileges**.

4. Password Reset Does Not Resolve Lockout

- Check if the user has logged into **other devices or applications** with old credentials.
- Ensure all cached credentials are updated.
- Verify that the account is **replicated correctly across all domain controllers**.

5. Suspicious or Unauthorized Lockouts

- Escalate to **Network/Security Team** immediately.
- Review logs for **failed login attempts** from unknown IP addresses or devices.
- Consider **temporary account suspension** or further investigation of compromised accounts.

6. General Recommendations

- Always document all actions and findings in the ticketing system.
- Regularly update **tools, scripts, and administrative access** to ensure smooth operations.
- Schedule periodic reviews of recurring lockouts to implement **preventive measures**.

9. Escalation Procedure

This section defines when and how to escalate Active Directory account lockout issues that cannot be resolved through standard procedures.

When to Escalate:

- Account remains locked **after following standard unlocking steps**.
- Lockout **source is unknown or cannot be identified** using Event Viewer and LockoutStatus tool.
- Evidence of **suspicious or potentially malicious activity** (multiple failed logins from unknown IPs or devices).
- Repeated lockouts affecting **multiple users** indicating a systemic or network-wide issue.
- Failure of tools or access permissions preventing resolution.

Who to Contact:

- **Senior System Administrator** – primary escalation point for unresolved lockouts.
- **Network / Security Team** – for suspected security incidents.
- **IT Manager / Incident Response Team** – if the issue affects critical systems or multiple accounts.

How to Log the Escalation:

- Update the **ticketing system** with:
- ✓ Description of issue and steps already taken
- ✓ User account(s) affected
- ✓ Caller Computer(s) or affected devices
- ✓ Evidence from logs or tools (screenshots, event IDs)
- ✓ Contact information of staff handling the escalation
- Tag the ticket with **"Escalated"** and notify the appropriate personnel via **email or messaging system**.
- Maintain documentation for **audit, compliance, and review purposes**.

Additional Notes:

- Always communicate escalation promptly to **reduce downtime and risk**.
- Follow **organization's incident response and security policies** for escalated events.
- Escalation does not replace **documentation**; all findings must still be logged.

10. References / Supporting Materials

This section provides sources and materials that support the procedures in this SOP.

1. Official Documentation:

- Microsoft Active Directory Documentation
- Event ID 4740 – Microsoft
- Active Directory Module for Windows PowerShell

2. Internal Policies and Procedures:

- Organization-specific **IT Security Policy** for account management
- **Incident Response Policy** for reporting suspicious lockouts
- **Escalation Policy** and contact list for IT/Security teams

3. Standards and Compliance:

- **ISO/IEC 27001:2013** – Information Security Management System (ISMS)
- **ISO/IEC 27002:2013** – Security controls and guidance for AD and network security

4. Supporting Tools and References:

- **LockoutStatus Tool** – for identifying locked accounts across domain controllers
- **Event Viewer** screenshots or example filters
- **Sample PowerShell commands** used for unlocking accounts and verifying logs

5. Optional / Appendices:

- Sample **ticket entries** for account lockout incidents
- Diagrams of **AD structure and lockout flow**
- Example **escalation contact table**

11. Revision History

Version	Date	Author	Changes Made / Description
1.0	2026-03-11	Nabin Shrestha	Initial draft – complete SOP for AD User Account Lockout Resolution

Note: Update this table whenever the SOP is revised to track changes, authors, and dates.

FLOWCHARTS

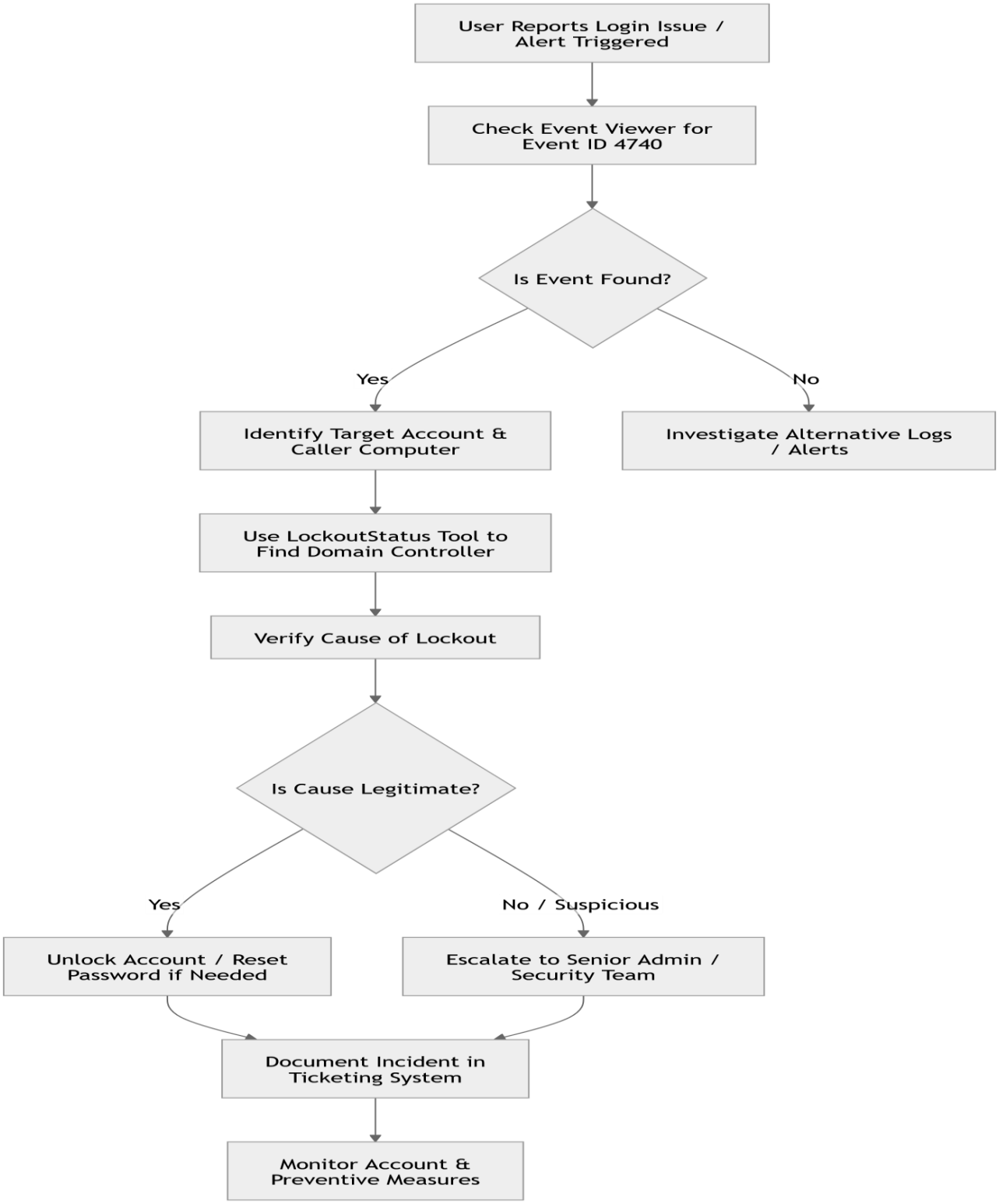

Figure: AD User Account Lockout Resolution Flowchart

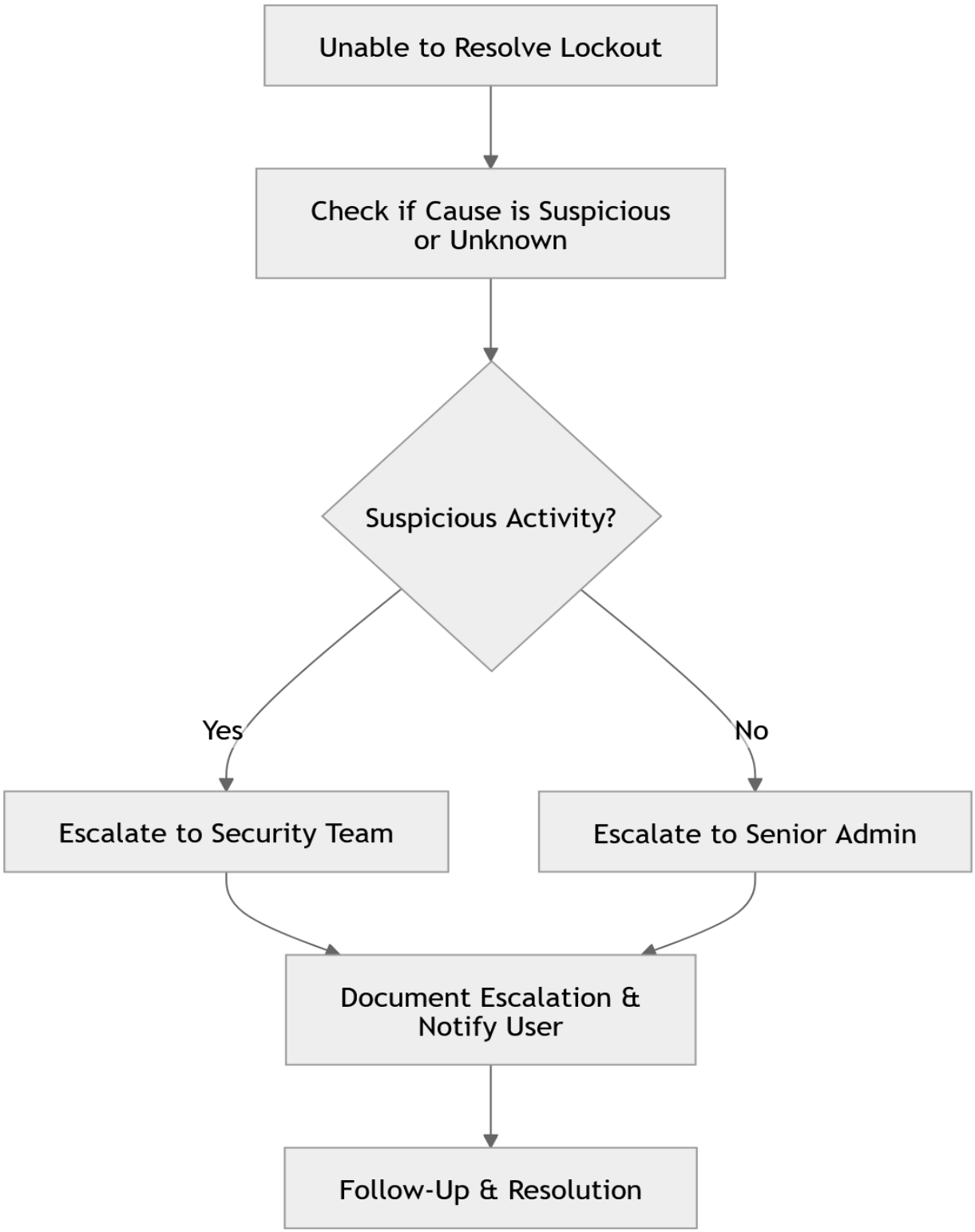

Figure: Escalation Flowchart

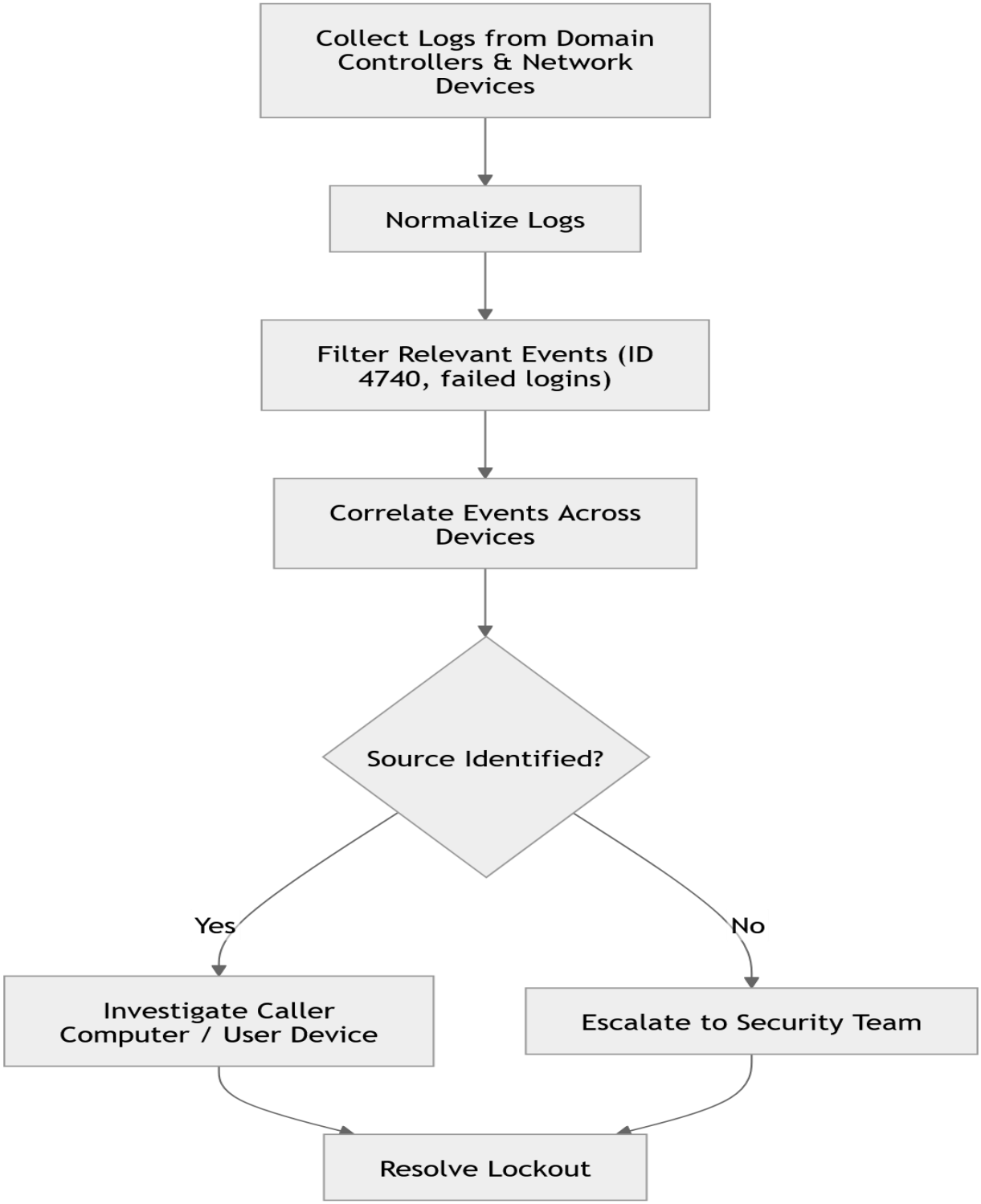

Figure: Event Correlation / Log Analysis Flow

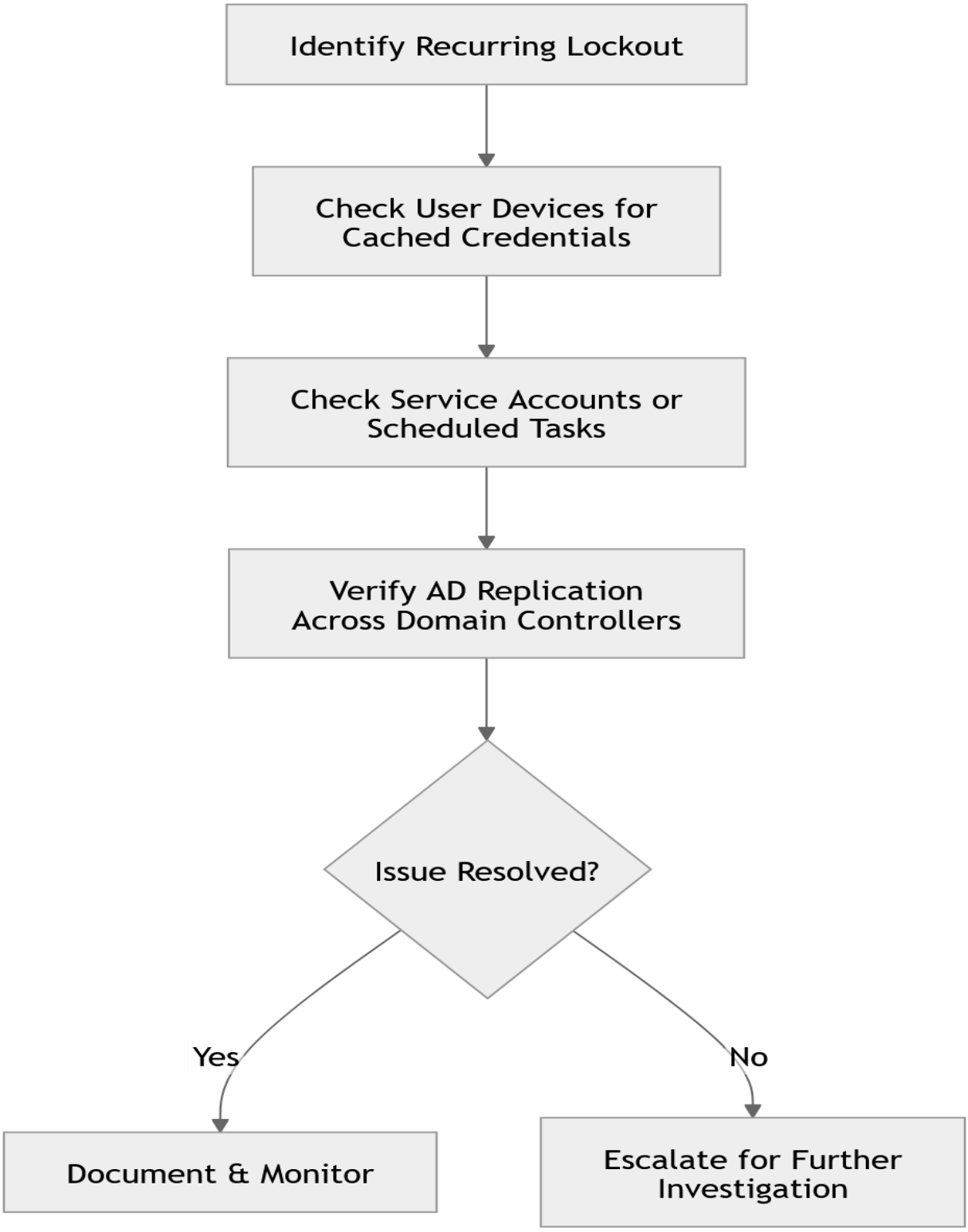

Figure: Preventive Measures / Recurring Lockout Handling